

Healthcare Workforce Management Platform - Technical Architecture

1) Core Business Modules

- Job Order Management
- Candidate Workflow
- Vendor Management
- Onboarding
- Timesheet
- Invoicing

2) High-Level Architecture Diagram (Text)

Users (Hospitals/MSP/Vendors)

- > Route53 -> CloudFront + WAF + Shield
- > React SPA (S3)
- > API Gateway -> ALB -> EKS (Spring Boot Microservices)
 - > Services: Job Order, Candidate Workflow, Onboarding, Timesheet, Invoicing, Vendor Mgmt, Notification
 - > Aurora PostgreSQL (per-service DB/schema)
 - > ElastiCache Redis
 - > S3 (documents and invoice artifacts)
 - > OpenSearch (analytics/search)
 - > MSK Kafka (real-time events: job broadcast to vendors)

3) Cloud Technologies (Real Tools)

- Frontend: ReactJS, Amazon S3, Amazon CloudFront
- API & Routing: Amazon API Gateway, Application Load Balancer
- Microservices Platform: Java, Spring Boot, Amazon EKS
- Event Streaming: Apache Kafka on Amazon MSK
- Databases: Amazon Aurora PostgreSQL (+ read replicas)
- Cache: Amazon ElastiCache for Redis
- Search/Analytics: Amazon OpenSearch Service
- Files: Amazon S3
- Identity: Amazon Cognito or Okta (OIDC/OAuth2, JWT)
- Observability: CloudWatch, Prometheus, Grafana, OpenTelemetry, X-Ray
- CI/CD: GitHub Actions + Argo CD (or AWS CodePipeline)

4) Security Controls

- TLS 1.2+ at edge and service communication
- WAF managed rules, DDoS protection via Shield
- Private subnets for services/data, strict Security Groups
- IAM least privilege, IRSA for pod-level AWS access
- Secrets in AWS Secrets Manager; keys in AWS KMS
- Encryption at rest: Aurora, S3, MSK, EBS
- OAuth2/OIDC authN and RBAC/ABAC authZ
- Audit and threat detection: CloudTrail, GuardDuty, Security Hub

5) Scalability, Performance, Fault Tolerance

Scalability:

- HPA on CPU/memory/Kafka lag; cluster autoscaling
- Kafka topic partition scaling for parallel consumers
- CDN and Redis offload for high read throughput

Performance:

- Event-driven async workflows reduce synchronous blocking
- Query optimization, indexing, connection pooling
- Separate OLTP (Aurora) and analytics/search (OpenSearch)

Fault Tolerance:

- Multi-AZ EKS/MSK/Aurora
- Kafka replication factor 3, retries, idempotent consumers
- DLQ topics + backoff retries + circuit breakers
- Outbox pattern for reliable event publishing
- Automated backups, PITR, optional cross-region DR failover